Stewart Tabulated Environment Gate

No-jet Nectar stability check, 2026-05-05

2026-05-05

Stewart Tabulated Environment Gate, 2026-05-05

Post-meeting correction, `2026-05-07`: the environment-gate results remain valid, but the "next gate" and jet-duty-cycle unit interpretation below were superseded by the 2026-05-06 supervisor meeting. `JET_END_TIME` and `JET_ACTIVE_TIME` are physical Myr values in the user parameter block; PLUTO core controls such as `tstop` and output intervals are the values that use PLUTO code time.

Summary

The confirmed Stewart radially averaged environments passed the first no-jet Nectar stability gate on `2026-05-05`.

Stas' clarification is now treated as binding for science runs: use the tabulated radius profiles, not the full 3D grid environments, because the full 3D profiles have too much spurious motion for reliable energetics.

Inputs

The tested files were extracted from the ignored `env/environments.zip` archive and staged to `nectar-vm`:

Science role	Stewart environment	Extracted asset
`C-HI`, `C-LO`	cluster `002_003`	`tabulated_cosmo_002003.csv`
`G-LO`	group `002_031`	`tabulated_cosmo_002031.csv`

The local MacBook remained the control plane. The simulations ran on `nectar-vm` after resizing to `c3.3xlarge`.

Gate Setup

The tests were rendered with `scripts/science/render_stewart_environment_gate_tests.py`, using the extracted cluster and group CSVs, the current `jet-setup` environment verifier, and the current `jet-setup` makefile and `definitions.h` templates.

Each rendered gate uses:

- HD, 3D Cartesian, vector body force
- no jet: `JET_END_TIME = 0`, `JET_ACTIVE_TIME = 0`
- no particles: rendered `definitions.h` sets `PARTICLES = NO`
- grid: `50 x 50 x 50`, uniform `[-100, 100] kpc` in each dimension
- run duration: `300 Myr`
- PLUTO code-time conversion: `1 Myr = 306.601393788`
- outputs near `0`, `100`, `200`, and `300 Myr`

Results

Environment	Max central density drift	Max central pressure drift	Max Mach	Max velocity	Gate
cluster 002_003	0.136%	0.226%	0.0242	10.6 km/s	pass
group 002_031	0.284%	0.473%	0.0330	8.07 km/s	pass

The `jet-setup` verifier passed all outputs for both environments. These are smoke-scale stability gates, not final production-resolution validation.

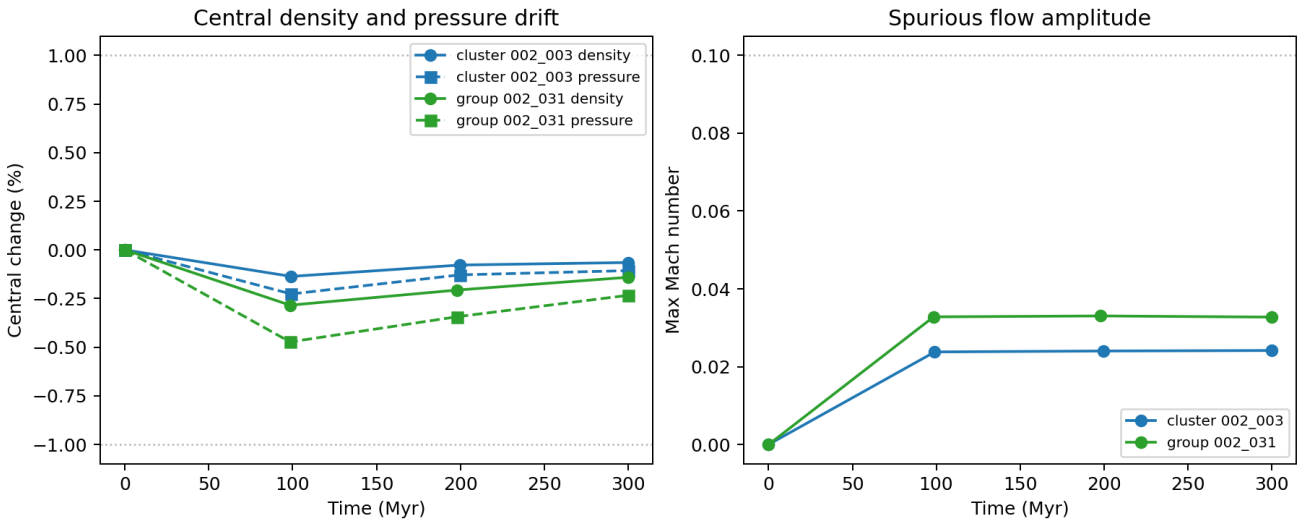


Figure 1: Stewart environment gate time series

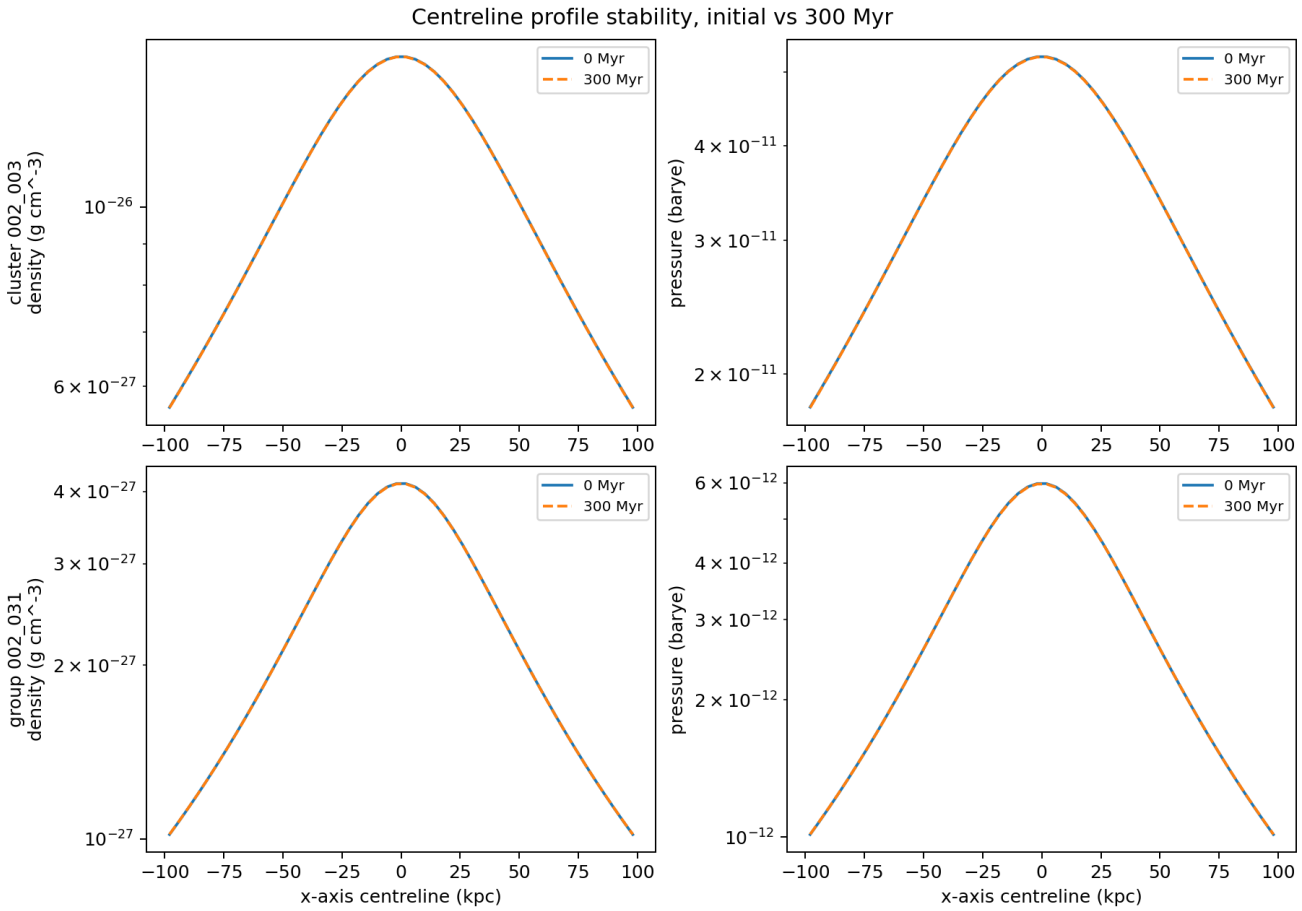


Figure 2: Stewart environment gate profiles

The plotted metrics are stored as `stewart_environment_gate_metrics.csv` in the matching asset directory.

Implementation Notes

Two small operational fixes came out of the gate:

- `scripts/science/render_stewart_environment_gate_tests.py` now renders a hydro-only `definitions.h` and a particle-free `local_make`, because the current `jet-setup` source template otherwise keeps `PARTICLES_LP` enabled.
- `scripts/science/render_three_case_configs.py` was later corrected after the supervisor meeting so `JET_END_TIME` and `JET_ACTIVE_TIME` remain physical Myr values; only `tstop`, output intervals, and checkpoint intervals are converted to PLUTO code time.

During the run, `nectar-vm` SSH was switched from fragile socket activation to persistent `ssh.service`; this is recorded in `docs/nectar_vm_setup.md`.

Next Gate

The environment asset blocker is now cleared. After the supervisor meeting, the next science gate is the production-like `C-HI` high-power cluster package using:

- cluster tabulated environment `002_003`
- `JET_PWR = 1e45 erg/s`
- `JET_SPD = 0.1`
- `JET_OA_PRIMARY = JET_OA_SECONDARY = 15 deg`
- particles enabled for the nominal science run
- first target duration about `50 Myr`
- early outputs at about `1 Myr` cadence
- double restart checkpoints every `2-5 Myr` and at walltime stop
- HDF5/MPI validation on `kunanyi`

Hydro-only `C-HI` is now a narrow debugging fallback, not the planned first science gate.